

Topological Reachability Graph Construction & Sliding Lookahead Dijkstra (0% Map Cheating)

1 Offline Landmark Embeddings

Replay Buffer $\mathcal{D}_{\text{offline}} = \{s_t\}_{t=1}^{|\mathcal{D}|}$
 Uniform Sampling: $\mathcal{V} = \{s_i\}_{i=1}^N$ ($N = 1000$)

Backward Latents $z_i \in \mathbb{S}^{d-1}$

$$z_i = \text{norm}(B(s_i)) = \frac{B(s_i)}{\|B(s_i)\|_2} \cdot \sqrt{d}$$

Frozen B -representation ($d = 128$)

Coverage: Dense topological skeleton over state space $\mathcal{S}$ without map prior.

2 FB Reachability Matrix

Vectorized Inner Product $\mathbf{R} \in \mathbb{R}^{N \times N}$

$$\mathbf{R}_{ij} = F(s_i, z_j)^\top z_j \approx \frac{dM_{s_i z_j}^{\pi z_j}}{d\rho}(s_j)$$

JIT-compiled in 45k vectorized chunks

Quasi-Metric Cost Function

$$c_0(s_i, s_j) = \max(0, -\log \frac{\mathbf{R}_{ij}}{M_{\max}})$$

where $M_{\max} = \max_k (F(s_k, z_k)^\top z_k)$

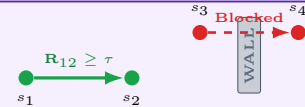
Multiplicative path probabilities map into additive weights: $\prod P_{ij} \leftrightarrow \sum c_{ij}$.

3 Dynamic Edge Filtering

Dual Feasibility Constraints

$$c(s_i, s_j) = \begin{cases} c_0(s_i, s_j), & \text{if } d_{ij} \leq R_{\max} \wedge \mathbf{R}_{ij} \geq \tau \\ \infty, & \text{otherwise} \end{cases}$$

Radius $R_{\max} = 3.5$ m, Threshold $\tau = 35.0$



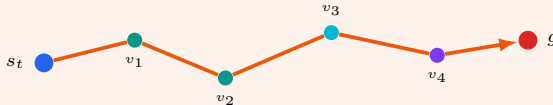
Zero Map Cheating: Rejects Euclidean shortcuts through walls dynamically.

4 Global Dijkstra Shortest Path Search on $G = (V, E)$

Shortest Path Dijkstra: $\mathcal{P}^* = \arg \min_{\mathcal{P}} \sum_{k=0}^{K-1} c(v_k, v_{k+1})$

Boundary Conditions: $v_0 \approx s_t$, $v_K \approx g$

Predecessor Matrix: $\mathbf{\Pi} \in \mathbb{R}^{N \times N}$ in $O(V^2)$



Feasible Multi-Hop Corridor Trajectory $\mathcal{P}^* = (v_0, v_1, \dots, v_K, g)$

5 Continuous Sliding Lookahead Sampling ($L = 2.6$ m)

Polyline Distance Accumulation ($L = 2.6$ m):

$$k^* = \arg \min_m \|s_t - v_m\|_2, \quad j^* = \min \left\{ j > k^* \mid \sum_{m=k^*}^{j-1} d_{m, m+1} \geq L \right\}$$

Lookahead Waypoint Sequence Output:

$$\mathcal{P}_{\text{lookahead}} = [w_1, w_2, \dots, w_K, g] \quad \text{where } w_1 = v_{j^*} \implies z_t = B(w_1) \in \mathbb{S}^{d-1}$$

Eliminates stop-and-go jitter, corner oscillations, and wall snagging.

Result: Success Rate $93.3 \pm 1.4\%$ on antmaze-medium-navigate-v0 (1.02 ms/step).

Offline Graph: $O(V^2)$ once (0.8 s)

Runtime Query: $O(1) + O(K)$ (1.02 ms)

Zero Map Prior: 100% Data-Driven